

ALGAZY OSTROV, Kazakhstan

WMO: 366860

Lat: 46.543N

Lon: 76.862E

Elev: 349

StdP: 97.20

Time Zone: 6.00 (E06)

Period: 02-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.7	-24.6	-29.9	0.2	-26.3	-27.9	0.3	-24.4	14.3	-14.3	12.0	-12.2	1.7	50	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	33.2	19.3	31.4	19.3	29.9	19.1	22.3	28.7	21.2	27.6	20.5	27.1	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.4	26.4	19.0	14.4	25.6	18.0	13.5	25.0	67.5	28.7	63.6	27.8	60.9	27.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.2	9.3	-28.9	37.0	2.8	2.0	-30.9	38.4	-32.5	39.5	-34.1	40.6	-36.1	42.1
			-29.1	24.0	2.8	1.5	-31.1	25.1	-32.7	25.9	-34.2	26.8	-36.2	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-13.8	-11.6	-1.6	9.2	17.2	23.2	24.8	23.8	17.5	9.6	1.4	-8.9	(6)
(7)		DBStd	14.40	5.78	6.51	6.15	4.77	3.76	3.24	2.83	3.09	4.03	3.82	4.53	6.44	(7)
(8)		HDD10.0	2677	739	606	361	70	1	0	0	1	53	258	587		(8)
(9)		HDD18.3	4491	998	839	619	276	67	4	1	2	62	270	508	846	(9)
(10)		CDD10.0	1818	0	0	0	45	223	396	459	427	227	42	1	0	(10)
(11)		CDD18.3	591	0	0	0	1	31	149	201	170	38	1	0	0	(11)
(12)		CDH23.3	4572	0	0	0	11	209	1162	1717	1284	188	1	0	0	(12)
(13)		CDH26.7	1485	0	0	0	1	51	377	587	424	44	0	0	0	(13)
(14)	Wind	WSAvg	3.9	3.0	2.9	3.6	3.8	4.3	4.2	4.3	4.5	4.2	4.4	3.5	(14)	
(15)	Precipitation	PrecAvg	125	9	9	9	11	14	9	17	6	9	15	12	(15)	
(16)		PrecMax	226	25	23	24	32	46	47	124	44	37	26	30	38	(16)
(17)		PrecMin	70	0	1	1	0	1	0	1	0	0	0	2	0	(17)
(18)		PrecStd	42	7	6	7	9	10	10	25	9	8	8	8	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	2.8	13.2	24.9	30.6	34.4	35.4	31.2	21.7	12.0	3.4	(19)	
(20)			MCWB	0.5	1.1	7.2	13.3	17.2	18.7	20.2	19.9	17.1	12.3	9.4	2.0	(20)
(21)		2%	DB	-1.0	1.0	9.8	20.9	28.0	32.1	33.1	32.6	27.4	18.6	10.0	1.4	(21)
(22)			MCWB	-1.9	0.0	5.3	12.0	16.4	18.9	19.4	20.0	16.9	12.0	7.9	0.2	(22)
(23)		5%	DB	-3.0	-0.6	7.4	18.0	25.6	30.4	31.4	30.7	25.2	16.7	8.5	0.4	(23)
(24)			MCWB	-3.7	-1.7	4.0	11.0	15.5	19.0	19.3	19.7	16.0	11.5	6.5	-0.7	(24)
(25)		10%	DB	-5.5	-2.4	5.4	15.9	23.5	28.7	29.8	29.0	23.5	15.0	7.1	-0.7	(25)
(26)			MCWB	-6.3	-3.4	2.8	10.0	14.8	18.5	19.4	19.2	15.3	10.5	5.1	-1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.4	8.1	14.7	19.1	23.6	23.9	24.6	19.7	14.5	10.2	2.2	(27)
(28)			MCDWB	1.0	2.6	12.0	21.7	26.8	30.2	30.9	31.3	27.0	18.2	11.6	2.8	(28)
(29)		2%	WB	-1.7	-0.1	5.9	13.1	17.7	21.4	22.2	22.2	18.2	13.1	8.2	0.5	(29)
(30)			MCDWB	-1.0	1.0	9.3	18.9	25.0	27.5	28.7	28.8	25.1	17.0	9.5	1.3	(30)
(31)		5%	WB	-3.7	-1.6	4.2	11.8	16.7	20.5	21.4	20.9	17.2	11.9	6.7	-0.6	(31)
(32)			MCDWB	-3.0	-0.6	6.9	16.9	23.3	27.1	27.8	27.4	23.7	15.6	8.4	0.2	(32)
(33)		10%	WB	-6.2	-3.4	2.9	10.5	15.6	19.7	20.7	20.1	16.1	10.9	5.3	-1.7	(33)
(34)			MCDWB	-5.5	-2.4	5.3	15.1	21.9	26.7	27.2	27.0	22.5	15.0	6.9	-0.7	(34)
(35)	Mean Daily Temperature Range	MDBR	7.9	8.6	8.2	9.5	9.7	9.4	8.8	9.1	8.4	6.8	5.0	6.4	(35)	
(36)		5% DB	MCDBR	8.3	7.8	10.0	12.3	12.5	11.1	10.7	11.2	10.4	9.2	6.1	5.5	(36)
(37)			MCWBR	7.9	7.3	6.8	5.9	4.8	4.1	4.2	4.8	5.1	5.5	4.6	5.0	(37)
(38)		5% WB	MCDWB	7.9	7.7	9.5	10.7	11.1	9.6	9.2	9.9	9.4	8.1	5.9	5.3	(38)
(39)			MCWBR	7.6	7.4	6.8	5.7	5.3	4.7	4.8	5.3	5.5	5.7	4.6	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.353	0.401	0.443	0.447	0.447	0.453	0.407	0.371	0.373	0.351	0.318	(40)	
(41)		taud	2.036	1.989	2.043	2.015	2.057	2.122	2.130	2.239	2.293	2.192	2.143	2.072	(41)	
(42)		Ebn at Noon	725	782	807	809	821	823	812	837	836	760	690	678	(42)	
(43)		Edh at Noon	100	129	143	162	161	152	149	128	111	105	90	86	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.58	2.51	3.88	5.71	7.12	7.46	7.13	6.56	5.10	3.10	1.73	1.10	(44)	
(45)		RadStd	0.25	0.24	0.27	0.28	0.23	0.35	0.40	0.22	0.20	0.24	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page